Case Study: DeepMind's AlphaGo

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1. Introduction

- **AlphaGo** is an artificial intelligence program developed by **DeepMind Technologies** (a subsidiary of Google).
- It was designed to play the ancient board game **Go**, which is significantly more complex than chess due to its vast search space and deep strategic depth.
- AlphaGo made history in **March 2016** by defeating **Lee Sedol**, one of the world's top Go players, with a score of 4–1.

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2. Why AlphaGo Was Revolutionary

- **Go Complexity**:
  - The number of possible board configurations is estimated at 10^{170}, making brute-force search (**Brute-Force Search** (also called **exhaustive search**) is a

problem-solving technique that involves trying **all possible solutions** until the correct one is found) infeasible.

- Traditional game-playing algorithms (like those in chess engines) fail due to Go's complexity.

- **AI Breakthrough**: AlphaGo combined **deep neural networks** and **reinforcement learning** to achieve superhuman performance.

    (**Reinforcement Learning** is a type of **machine learning paradigm** where an **agent learns by interacting with an environment**, receiving feedback in the form of **rewards or penalties** for its actions, and aims to maximize its cumulative reward over time.)

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3. Technology Behind AlphaGo

AlphaGo’s success is due to its innovative architecture and algorithms:

1. **Deep Neural Networks**:
    - **Policy Network**: Suggests the next best moves by predicting probabilities.

- **Value Network**: Evaluates a board position to estimate the chance of winning.

2. **Monte Carlo Tree Search (MCTS)**:
   - Explores possible moves, guided by the policy and value networks.

3. **Reinforcement Learning**:
   - Trained by playing millions of games against itself (self-play).
   - Initially trained on 30 million human expert moves (supervised learning), then improved via reinforcement learning.

4. **Hardware**:
   - Used **Google's TPUs (Tensor Processing Units)** for fast computations.

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4. Key Milestones

- **October 2015**: AlphaGo defeated European champion **Fan Hui** (5–0).
- **March 2016**: Defeated world champion **Lee Sedol** (4–1).

- **May 2017**: AlphaGo defeated the top-ranked player **Ke Jie** (3–0).
- **AlphaGo Zero** (2017): A newer version trained **without human data** — it learned entirely by self-play, surpassing the original AlphaGo in 40 days.

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5. Impact of AlphaGo

- **AI Advancement**: Showed that deep reinforcement learning could master complex tasks previously thought impossible for machines.
- **New Research Directions**: Inspired the development of AlphaZero, which mastered chess, shogi, and Go from scratch.
- **Cross-Industry Influence**:
  - Techniques from AlphaGo are now applied in **protein folding** (AlphaFold), robotics, and other domains.

(**Protein Folding** is the process by which a linear chain of amino acids (a polypeptide) folds into a **specific three-dimensional structure** that determines the protein's function in a biological system.)

- **Cultural Impact**: AlphaGo vs. Lee Sedol match attracted millions of viewers, demonstrating AI's potential to the public.

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6. Lessons Learned

- **Human + AI Synergy**: Human players learned new strategies from AlphaGo’s unconventional moves.
- **Self-Play is Powerful**: Reinforcement learning from self-play can surpass human knowledge.
- **Scalability**: AI models benefit significantly from powerful computing resources and large-scale data.

- **The board game go**

  Initially the board is bare, and players alternate turns to place one stone per turn. As the game proceeds, players try to link their stones together into "living" formations (meaning that they are permanently safe from capture), as well as threaten to capture their opponent's stones and formations.

- **Protein Folding – AlphaFold**

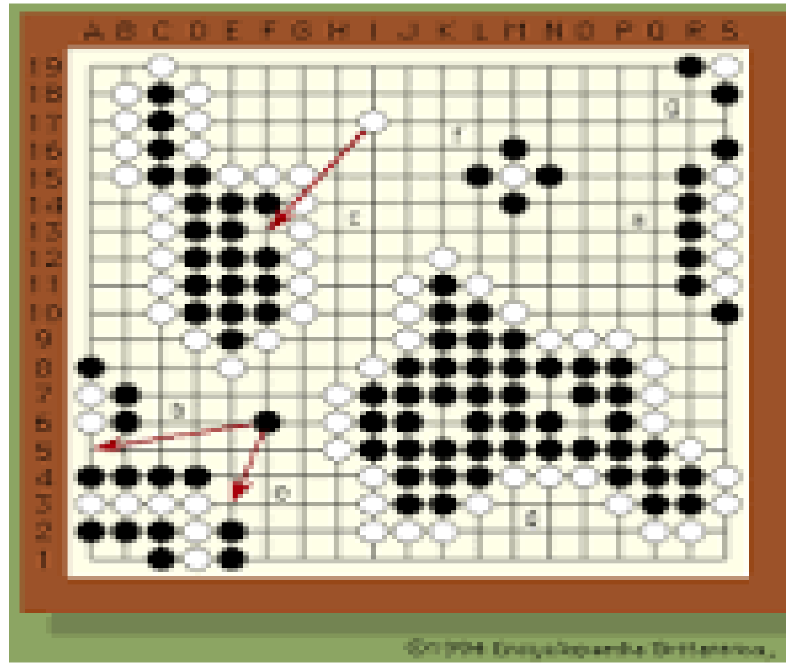
- **Problem**: Predicting the 3D shape of a protein from its amino acid sequence is extremely difficult and traditionally required expensive lab experiments.

- **Solution**: Using **deep learning with similar ideas to AlphaGo** (policy/value networks and learning from data), DeepMind developed **AlphaFold**.

- **Impact**:

    - AlphaFold can predict protein structures with near-laboratory accuracy.

    - It has solved structures for over **200 million proteins**, accelerating drug discovery and medical research.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